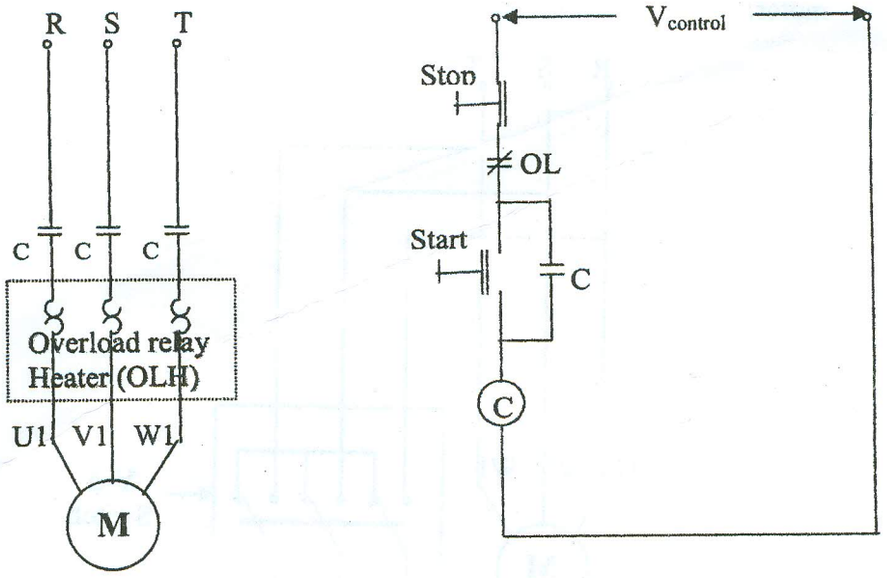
Laboratory job 18

Across the line motor start

The diagram shows a typical full-voltage magnetic starter equipped with three thermal overloads with start-stop buttons.

1. Explain how the circuit shown below operates.
2. Sometimes, it is advisable to have a pilot light on the control circuit. Draw such a control circuit with the pilot light ON when the motor runs and a pilot light OFF when the motor stops.
3. Construct both the power and control circuit diagrams on the board. The instructor must check the circuit before you apply power.
4. What is the main problem with this type of starting method on circuit breaker, over load relay, magnetic starter and on the motor itself?
5. Can large motors (with power rating greater than 10hp) be started using this method?



Power Circuit **Control Circuit**

Fig 18 Across the line motor starting method

Question

Repeat Q2 so that the pilot light turns OFF when the motor runs and ON when it stops.